

LeetCode 977: Squares of a Sorted Array

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 **Interview Appearances:** Deutsche Bank, Whatfix

1. Approach & Logic Notes

Approach: *nums array already sorted so kaayde se largest element peeche wala hoga but since we are squaring agar aage koi negative he jiska magnitude peeche waala se bada he to uska square bhi bada hoga , hence we use 2 pointers , one at begin one at end*

2. Implementation (C++)

```
class Solution {
public:
    vector<int> sortedSquares(vector<int>& nums) {

        //approach: nums array already sorted so kaayde se largest
        element peeche wala hoga but since we are squaring agar aage koi
        negative he jiska magnitude peeche waala se bada he to uska square bhi
        bada hoga , hence we use 2 pointers , one at begin one at end

        int n = nums.size();
        vector<int> ans(n);
        int i = 0;
        int j = n - 1;

        int idx = n - 1;

        while(i <= j){
            if(abs(nums[i]) > abs(nums[j])){
                ans[idx] = nums[i] * nums[i];
            }
            else{
                ans[idx] = nums[j] * nums[j];
            }
            idx--;
            if(i < j){
                i++;
                j--;
            }
        }
    }
};
```

```
        idx--;  
        i++;  
    }  
    else{  
        ans[idx] = nums[j] * nums[j];  
        idx--;  
        j--;  
    }  
}  
  
return ans;  
}  
};
```